RCLPG — Light Probe Generator v2.0

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1. Overview

RCLPG is an automated Light Probe placement tool for Unity. It generates probes inside a configurable volume, snapping them to terrain surfaces and thinning them out with height for optimal performance.

Features

- Optimized raycast pipeline for fast generation
- Probes hug terrain geometry and avoid all colliders
- Adaptive density control — one slider to thin out probes with height
- Volume follows the object transform — full rotation, position and scale support
- Real-time preview — changes visible instantly in Scene View
- Zero probes inside walls, floors, or ceilings

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2. Installation

1. Import the RCLPG package into your Unity project.
2. Locate the `RCLPG/Scripts` folder in your `Assets` directory.
3. In the Hierarchy, create an empty GameObject (right-click > **Create Empty**).
4. With the GameObject selected, click **Add Component** and search for **RCLPG**.
5. A `LightProbeGroup` component will be automatically attached.

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3. Quick Start

Follow these steps to generate your first probe layout:

Step 1 — Position the Volume

Place the GameObject in the scene where you want probes to appear. The red wireframe cube shows the probe volume.

Step 2 — Set the Volume Size

In the Inspector, adjust **Volume Size** (X, Y, Z) to cover the area where you need light probes.

Step 3 — Configure Spacing

Set **Horizontal** and **Vertical** spacing values to control base probe density. Smaller values = denser probes.

Step 4 — Set Ground Tags

In the Raycast section, add the tags of your terrain/ground objects to **Ground Tags** (default: `Ground`). Make sure your ground objects have matching tags.

Step 5 — Generate

Click the **Generate** button. Probes will appear as blue spheres in the Scene View, starting from the terrain surface upward.

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4. Parameters Reference

Volume

PARAMETER	TYPE	DEFAULT	DESCRIPTION
Volume Size	Vector3	100, 100, 100	Dimensions of the probe volume in meters
Volume Center Offset	Vector3	0, 0, 0	Offset of the volume center relative to the GameObject position

Spacing

PARAMETER	TYPE	DEFAULT	DESCRIPTION
Horizontal	float	5	Distance between probes on the XZ plane (meters)
Vertical	float	2	Distance between probes along the Y axis (meters)
Ground Clearance	float	0.5	Offset from terrain surface to avoid placing probes inside geometry
Probe Clearance Radius	float	0.3	Minimum safe distance from any collider. Probes inside colliders are skipped
Adaptive Spread	float (0–1)	0	Controls how probes spread out with height. 0 = uniform grid, 1 = maximum thinning

Raycast

PARAMETER	TYPE	DEFAULT	DESCRIPTION
Raycast Layers	LayerMask	Everything	Layers that raycast will detect for surface finding
Ignore Tags Filter	bool	false	When enabled, ignores Ground Tags and uses any collider as surface
Ground Tags	string[]	Ground	Tags used to identify terrain. Probes start from the surface of matching objects

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5. Usage Tips

Avoiding Probes in Geometry

If probes still appear inside objects, increase **Probe Clearance Radius** or add those objects' layers to **Raycast Layers**.

Performance vs. Quality

- Use smaller **Horizontal** / **Vertical** values for high-quality indoor areas
- Enable **Adaptive Spread** (0.3–0.6) for large outdoor areas to reduce probe count at height

- Larger volumes with Adaptive Spread at 0.5+ give the best quality/performance ratio

Working with Complex Terrain

Make sure all terrain pieces have the correct tag listed in **Ground Tags**. Use **Ignore Tags Filter** if you want probes to start from any collider regardless of tag.

Multiple Volumes

Create multiple RCLPG GameObjects for different areas (indoors, outdoors, dungeons, etc.) with individual settings per zone.

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6. Changelog

v2.0

- Faster generation — optimized raycast pipeline
- Higher accuracy — probes perfectly hug terrain and avoid all geometry
- Adaptive density — one slider to thin out probes with height
- Volume follows the object — full rotation, position and scale support
- Cleaner interface — simpler controls, less clutter
- Real-time preview — instant feedback as you tweak settings
- Zero probes inside walls, floors or ceilings — built-in collision detection